Fabric Cutting Machine



Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Jute Fabric Cutting Machine

What it is used for

A fabric cutting machine cuts jute or other approved textile materials into specified lengths and shapes for sewing, packaging, or production. Correct blade selection, fabric alignment, and secure handling help achieve accurate cuts while reducing injuries and material waste.

Main parts

- Machine frame and cutting table
- Cutting blade, knife, or rotary cutter
- Fabric guides and measuring scale
- Clamps or fabric holding mechanism
- Motor, drive system, and operating controls
- Blade guard, emergency stop, and dust extraction where fitted

Before you start

1. Confirm the fabric type, thickness, required dimensions, and cutting pattern.
2. Inspect the blade, guard, cutting table, power cable, controls, and emergency stop for damage.
3. Verify that the blade is suitable for jute fabric and is securely installed according to the manual.
4. Ensure the cutting table is clean, stable, and free from pins, tools, and other obstructions.
5. Spread and align the fabric without folds, excessive tension, or overlapping layers beyond machine capacity.
6. Wear appropriate eye protection, suitable footwear, and other required PPE; secure loose clothing and hair.

How to use

1. Switch off and isolate power before changing the blade, adjusting guides, or performing setup work.
2. Set the cutting length, width, guide position, and other required settings according to the cutting plan.
3. Secure the fabric using the approved holding method and confirm that the cutting path is clear.
4. Check that the blade guard and other safety devices are correctly positioned.
5. Switch on the machine and perform a test cut on a suitable fabric sample if required.
6. Feed or guide the fabric at the recommended speed without forcing it or placing hands near the blade.
7. Maintain alignment with the measuring scale or cutting guide throughout the cut.

8. Observe the cutting quality and stop the machine if the fabric shifts, the blade binds, or unusual vibration occurs.
9. Switch off the machine and wait for all moving parts to stop before repositioning fabric or removing offcuts.
10. Inspect the cut edges and dimensions against the production specification.
11. Repeat the cut only after correcting any alignment or setting issues.
12. At the end of the job, switch off and isolate power before removing scraps and cleaning the cutting area.
13. Record material usage, rejected pieces, and any machine faults according to workplace procedures.

⚠ Safety

- Only trained operators should use the fabric cutting machine.
- Keep fingers and hands away from the blade and cutting path; never guide fabric directly in front of an exposed blade.
- Do not operate the machine with a missing or damaged blade guard or disabled emergency stop.
- Isolate power before blade replacement, jam removal, or maintenance.
- Do not cut fabric that exceeds the machine's rated thickness or capacity.
- Keep the floor and work area free from loose fabric, threads, and scraps that may cause slips or entanglement.
- Use suitable dust control and respiratory protection if cutting generates significant fibres or dust.

Care & storage

- Remove fabric scraps, lint, and dust after shutdown using the approved cleaning method.
- Inspect blade sharpness, blade mounting, guards, guides, and clamps regularly.
- Replace worn or damaged blades with the correct approved type.
- Check drive components, power cables, and emergency-stop function according to the maintenance schedule.
- Arrange servicing if cut quality deteriorates, the motor overheats, or abnormal vibration occurs.

Common mistakes to avoid

- Using a dull or unsuitable blade for the fabric thickness.
- Failing to align or secure the fabric before cutting.
- Reaching near the blade while the machine is running or coasting.
- Ignoring damaged guards or continuing after the fabric becomes jammed.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.